

IMPROVING DEXTEROUS PIANO PLAYING POLICIES IN PIANOmime

DRL Course Project Group 28

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ABSTRACT

We study dexterous robotic piano playing in PianoMime, where two anthropomorphic Shadow Hands execute target key-press trajectories on an 88-key piano. We reproduce the original single-song specialist policies and the multi-song generalist policy, then evaluate targeted modifications for note-level F1. In the specialist setting, onset-aware reward shaping, light smoothness regularization, and residual factor calibration improve F1 on `Happy_8` from 0.8725 to 0.9192. In the generalist setting, a motion-weighted low-level diffusion fine-tune improves the five-clip mean F1 from 0.6150 to 0.6258. Smooth denoised diffusion training, goal side-channel conditioning, and reward-based rollout selection also produce positive gains. RPIM-based integration gives useful diagnostic evidence and shows unstable average gains on the selected five-clip benchmark. For the course bonus, we also report an arm-mounted Shadow Hand result: mounting both hands on UR10-style arms achieves no-clamp F1 0.8073 on `Happy_8` with a reference-based re-targeting and action-sequence optimization pipeline. The results highlight the role of note timing, smooth hand motion, goal conditioning, representation alignment, and embodiment constraints in dexterous piano playing.

1 INTRODUCTION

Robotic piano playing provides a compact benchmark for high-dimensional dexterous control. The agent controls two anthropomorphic robot hands, observes robot state and future note goals, and produces joint-level actions or residual corrections. The task requires spatial accuracy, contact timing, and multi-finger coordination. A policy can generate plausible finger motion and still lose note-level F1 through missed onsets, weak presses, or wrong-key contacts.

RoboPianist introduced simulated bimanual piano playing as a goal-conditioned reinforcement learning task with anthropomorphic hands and a full-size keyboard (Zakka et al., 2023). PianoMime extends this line by using internet demonstrations and paired MIDI files to train a generalist dexterous piano player (Qian et al., 2024). The PianoMime pipeline extracts fingertip motion from videos, converts MIDI into piano-roll goals, uses inverse kinematics to obtain nominal Shadow Hand trajectories, trains song-specific residual RL experts, and distills the resulting demonstrations into a two-stage generalist policy. The high-level policy predicts future fingertip trajectories from MIDI goals. The low-level policy maps fingertip targets and robot proprioception to hand actions.

Our project follows the Dexterous Piano Track. We first reproduce PianoMime baseline behavior for both single-song specialists and a multi-song generalist policy. We then study modifications motivated by the observed failure patterns in the reproduced setting. The specialist experiments focus on reward shaping, motion smoothness, and residual scaling. The generalist experiments focus on low-level diffusion training, explicit goal conditioning, stochastic rollout selection, and RPIM-based data integration. The report uses a direct organization: Section 2 introduces the task and prior systems, Section 3 presents the complete method design, Section 4 reports results, and Section 5 covers the arm bonus, and Section 6 discusses findings.

Our main contributions are as follows. First, we reproduce three PianoMime single-song specialists and a generalist policy on five unseen clips, with F1 curves, quantitative metrics, and videos. Second, we improve a specialist policy on `Happy_8` by combining onset-aware reward, light smoothness regularization, and residual factor calibration. Third, we compare multiple positive generalist

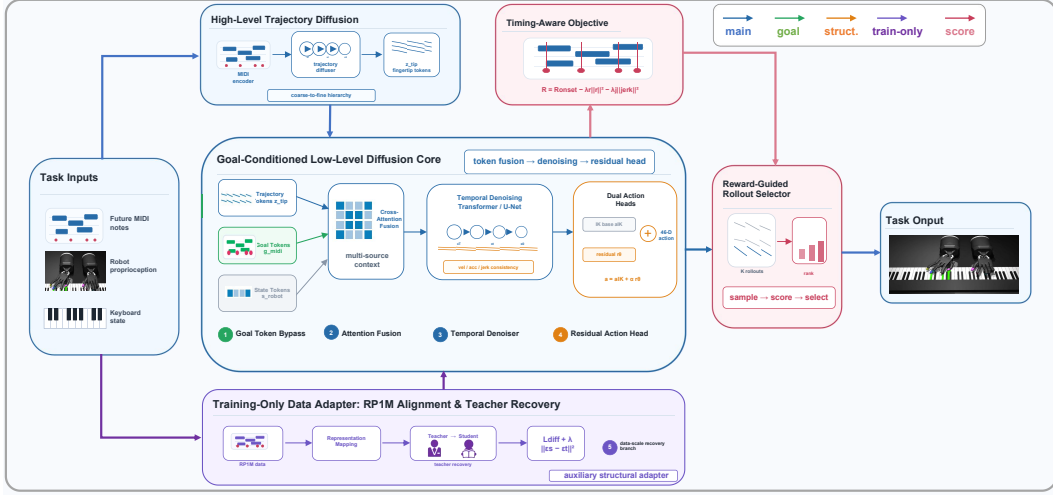


Figure 1: Architecture of our method. Future-note tokens bypass high-level trajectory prediction; multi-source attention fuses trajectory, goal, and robot tokens; a temporal denoising backbone generates smooth action sequences; and a calibrated residual head combines IK actions with learned corrections. Dashed purple paths indicate training-only RP1M alignment and teacher recovery.

modifications under the same five-clip benchmark and identify motion-weighted low-level diffusion fine-tuning as the best average improvement in the merged reader-facing comparison. Fourth, we report negative and diagnostic findings showing that naive smoothing, coarse song-level conditioning, and unregularized RP1M recovery can hurt note-level F1, highlighting the timing-sensitive nature of dexterous piano playing. Fifth, we implement a UR10-mounted Shadow Hand extension and analyze why a reference-based retargeting and action-sequence optimization pipeline is more reliable than direct arm-hand RL from scratch.

2 BACKGROUND AND RELATED WORK

2.1 ROBOTIC PIANO PLAYING AS AN RL PROBLEM

Each piano clip is an episodic control task. The environment contains an 88-key spring keyboard and two Shadow Hands. The policy observes proprioception, key states, and future note targets. The action is either direct hand control in the generalist setting or a residual correction in the specialist setting. The reward includes press success, mimic tracking, timing terms, and smoothness terms. The evaluation uses note-level precision, recall, and F1, plus qualitative performance videos.

This task combines several properties that make standard robot control difficult. The action space is high-dimensional because the policy controls two dexterous hands. The state evolves through contacts between fingertips and piano keys. The target trajectory is sparse in time because a note-level mistake often occurs near the press or release moment. A note may require preparation several frames before the visible key press. A small action change can move a fingertip away from the correct key or shift the press time. This property explains the emphasis on both quantitative F1 and video inspection in our experiments.

The single-song and generalist settings stress different parts of the problem. In the single-song setting, the policy can specialize to one fixed target trajectory and use residual control to refine a nominal motion. In the generalist setting, the policy must condition on future note goals and produce actions for unseen clips. This setting requires stronger goal-conditioned behavior because similar hand states can require different future actions under different local note patterns.

2.2 ROBOPIANIST, PIANOMIME, AND RP1M

RoboPianist trains simulated anthropomorphic hands to play piano pieces with deep reinforcement learning and provides an environment, benchmark tasks, evaluation metrics, code, and datasets (Zakka et al., 2023). It frames piano playing as a high-dimensional dexterous control problem with spatial precision, temporal precision, coordination, and planning. This formulation motivates the use of note-level metrics together with motion-quality inspection.

PianoMime builds a generalist agent from internet demonstrations (Qian et al., 2024). It uses paired piano videos and MIDI files. The video stream provides human fingertip motion. The MIDI stream provides task-level key-press goals. The data preparation stage extracts fingertip trajectories from videos and converts MIDI into piano-roll goals. Inverse kinematics then produces nominal Shadow Hand joint trajectories. Song-specific residual RL experts refine these nominal trajectories using rewards for key accuracy and fingertip-style tracking. The expert rollouts provide robot states, goals, fingertip trajectories, and actions for generalist policy distillation.

The PianoMime generalist policy has a hierarchical structure. The high-level policy predicts future fingertip trajectories from MIDI goals. The low-level policy converts predicted fingertip trajectories and proprioception into hand actions. At inference time, the agent uses MIDI goals and robot state as inputs. The original internet video is used during data construction and expert generation. This design links human demonstrations to robot control while keeping the final policy executable from task goals.

RP1M provides a large-scale motion dataset for bimanual robotic piano playing with around one million expert trajectories over around two thousand music pieces (Zhao et al., 2024). It uses optimal-transport-based fingering to automatically assign fingers to target keys, trains song-specific RL specialists, and collects their rollouts as robot state-action demonstrations. This dataset is closely related to our data-integration experiments because it provides robot demonstrations at a larger scale than a small set of PianoMime expert rollouts.

OmniPianist further scales this idea with more than two thousand specialists, RP1M++, and a Flow Matching Transformer for large-scale imitation learning (Chen et al., 2025). Flow Matching provides a generative modeling framework for learning continuous transformations from simple distributions to data distributions (Lipman et al., 2023). In our project, RP1M and OmniPianist motivate the attempt to use external robot piano-playing demonstrations for improving the PianoMime-style generalist policy.

2.3 DIFFUSION POLICIES FOR ACTION GENERATION

The PianoMime generalist policy follows a diffusion-based action generation pipeline. Diffusion models train a network to recover clean data from noisy data through a denoising objective (Ho et al., 2020). In robot learning, Diffusion Policy represents a visuomotor policy as a conditional denoising diffusion process over actions (Chi et al., 2023). This formulation is useful for high-dimensional continuous actions and stochastic action generation.

In the PianoMime low-level policy, the diffusion model predicts a 46-dimensional hand action conditioned on fingertip targets and proprioception. The model samples an action through iterative denoising. The denoised action estimate can be viewed as the clean action predicted by the model at a given diffusion step. This background is directly connected to our smooth denoised diffusion objective, which adds trajectory consistency terms to the denoised action estimate during training.

The stochastic nature of diffusion policies also motivates test-time rollout selection. Multiple samples from the same conditional diffusion policy can lead to different hand trajectories and different note-level outcomes. Reward-based rollout selection uses this stochasticity by sampling several rollouts and selecting the one with the best reward or F1 proxy.

2.4 PIANOMIME BASELINE POLICY STRUCTURE

Given a MIDI goal and robot state, the high-level diffusion model predicts future fingertip trajectories. Given the fingertip trajectory and proprioception, the low-level diffusion model predicts a 46-dimensional hand action. The default evaluation uses the official high-level checkpoint, low-

Table 1: Overview of insights and corresponding method components.

Observation	Method component	Setting
Note onset frames strongly affect note-level F1	Onset-aware reward	Specialist
Motion stability affects contact quality and execution robustness	Light smoothness regularization; smooth denoised diffusion objective	Specialist / Generalist
Residual strength controls the amount of learned correction	Residual factor calibration	Specialist
Future note patterns disambiguate low-level actions	Goal side-channel representation	Generalist
Stochastic diffusion rollouts have different execution quality	Reward-based rollout selection	Generalist inference

level checkpoint, and autoencoder checkpoint. In our reader-facing multi-task benchmark, all methods are evaluated on the same five unseen clips: `Alone_1`, `EyesClosed_1`, `Hope_1`, `SomewhereOnlyWeKnow_1`, and `NoTimeToDie_1`.

3 METHOD

Our method design is organized around five observations from the reproduced PianoMime setting. First, note onset frames have a strong influence on note-level F1, since a delayed or missed press at the beginning of a note can make an otherwise plausible motion fail the target key-press trajectory. This observation leads to onset-aware reward shaping in the specialist policy. Second, hand motion stability matters for both physical plausibility and generalist execution. A policy can use locally aggressive corrections to match notes, while such corrections may create unstable finger motion and fragile contacts. This observation leads to light smoothness regularization for the specialist policy and a smooth denoised diffusion objective for the generalist low-level policy. Third, the residual strength in the specialist policy controls the balance between the nominal IK action and the learned residual correction. This observation leads to residual factor calibration. Fourth, the same hand state can require different future actions under different upcoming note patterns. This observation leads to goal side-channel representation. Fifth, the diffusion policy is stochastic at inference, and different sampled rollouts can have different note-level quality. This observation leads to reward-based rollout selection.

3.1 SPECIALIST COMBINED METHOD

The specialist branch focuses on `Happy_8`. The reproduced baseline already generates recognizable piano-playing behavior, while remaining errors appear around note onsets, weak presses, and residual correction scale. We use a combined method with onset-aware reward, light smoothness regularization, and residual factor calibration.

The onset-aware reward increases the weight of frames where target notes begin. Let R_{task} denote the original task reward and let R_{onset} denote the additional reward on note onset frames. The specialist reward is written as

$$R = R_{\text{task}} + \lambda_{\text{onset}} R_{\text{onset}} - \lambda_{\Delta q} \|\Delta q\|_2^2 - \lambda_r \|r\|_2^2, \quad (1)$$

where r denotes the residual correction. The smoothness and residual penalties reduce unnecessary rapid corrections. The coefficients are kept light because dense notes still require fast finger movement.

The residual-control structure follows the PianoMime specialist pipeline. The policy starts from an IK-based nominal action and applies a learned residual correction. We calibrate the residual contribution at execution time:

$$a = a^{\text{IK}} + \alpha r, \quad (2)$$

where a^{IK} is the nominal IK action and α is the residual factor. This calibration controls the correction strength. A small residual factor keeps the policy close to the IK prior. A large residual factor increases the influence of learned corrections. The selected combined method uses the residual calibration reported by our single-song experiments.

3.2 SMOOTH DENOISED DIFFUSION OBJECTIVE

The generalist low-level policy predicts hand action chunks with a diffusion model. Local inconsistency inside an action chunk can produce unstable finger motion and unreliable contact. Direct test-time action smoothing can delay or suppress key contacts, so the smoothness constraint is applied during training on the denoised action estimate.

We keep the original diffusion noise-prediction loss and reconstruct the denoised clean action estimate $\hat{a}_0$ from the predicted noise. Auxiliary trajectory losses are applied on low-noise diffusion timesteps. The objective is

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{noise}} + 10^{-2} \|\Delta \hat{a}_0 - \Delta a_{\text{demo}}\|_2^2 + 5 \cdot 10^{-3} \|\Delta^2 \hat{a}_0 - \Delta^2 a_{\text{demo}}\|_2^2 \\ & + 10^{-3} \|\Delta^2 \hat{a}_0\|_2^2 + 10^{-3} \|\hat{a}_0 - a_{\text{demo}}\|_2^2. \end{aligned} \quad (3)$$

The first auxiliary term matches first-order action changes. The second term matches second-order jerk patterns from demonstrations. The third term penalizes unnecessary absolute jerk. The fourth term anchors the denoised action estimate to the demonstration action. These terms shape the denoised action chunk while keeping the diffusion sampling interface unchanged.

3.3 GOAL SIDE-CHANNEL REPRESENTATION

The low-level generalist policy receives proprioception and future fingertip targets. Similar hand states can require different actions when the upcoming notes change. We therefore add an explicit goal side-channel to the low-level conditioning. The side-channel provides future-note information to the action model and helps the policy use local note structure during action prediction. This component targets the goal-conditioning bottleneck of the generalist policy.

3.4 REWARD-BASED ROLLOUT SELECTION

The diffusion generalist policy can sample multiple rollouts under the same clip and checkpoint. These rollouts can differ in contact timing and note-level accuracy. We use reward-based rollout selection at inference time. The official high-level and low-level checkpoints are fixed. For each clip, multiple stochastic diffusion rollouts are sampled, and the rollout with the best reward or F1 proxy is selected. This component changes the inference selection procedure and keeps the model weights unchanged.

3.5 RP1M RECOVERY WITH TEACHER REGULARIZATION

The data-side branch studies whether RP1M-derived demonstrations can improve the PianoMime-style generalist policy. We evaluate two variants. The first variant uses RP1M-derived data to replace or initialize the low-level policy. The effective preprocessing includes point-major fingertip conversion and a learned linear mapping from RP1M representation to PianoMime-style representation. The second variant starts from the RP1M-derived low-level model and performs recovery on the original PianoMime low-level data with teacher regularization.

The recovery loss is

$$\mathcal{L}_{\text{recovery}} = \mathcal{L}_{\text{diffusion.noise.mse}} + \lambda \|\epsilon_{\text{student}} - \epsilon_{\text{teacher}}\|_2^2. \quad (4)$$

The first term trains the student on the PianoMime low-level diffusion task. The second term makes the student noise prediction close to the frozen official low-level teacher. This term guides the RP1M-derived student toward the official PianoMime low-level behavior distribution during recovery.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

We organize the experiments into baseline reproduction, specialist improvement, generalist improvement, and RP1M diagnostics. The single-song baseline clips are Happy_8,

Table 2: Single-song PianoMime baseline reproduction.

Clip	F1
Happy_8	0.8725
ImagineDragons_6	0.9158
LetMeDownSlowly_6	0.9289

Table 3: Specialist improvement on Happy_8.

Method	F1	$\Delta F1$
Baseline	0.8725	–
Onset + light smoothness + residual calibration	0.9192	+0.0467

ImagineDragons_6, and LetMeDownSlowly_6. The specialist improvement uses Happy_8 as the fixed comparison clip. The multi-song generalist benchmark uses five unseen clips from dataset/notes_test: Alone_1, EyesClosed_1, Hope_1, SomewhereOnlyWeKnow_1, and NoTimeToDie_1. All generalist methods are compared under the same five-clip protocol. We report precision, recall, F1, learning curves, and performance videos.

4.2 BASELINE REPRODUCTION

Table 2 reports the reproduced single-song specialist baselines. These results verify that the original PianoMime single-song setting runs successfully on the selected clips. The generalist baseline is reported together with the generalist modifications in Table 4.

4.3 SPECIALIST COMBINED METHOD

Table 3 reports the specialist result on Happy_8. The combined method improves F1 from 0.8725 to 0.9192, with a gain of 0.0467. The improved rollout reaches precision 1.0000 and recall 0.8895 in our summary. The baseline precision and recall are not included in the current single-song summary table, so the main table reports F1 and $\Delta F1$.

4.4 GENERALIST POLICY MODIFICATIONS

Table 4 reports the main generalist comparison on the five unseen clips. The motion-weighted low-level diffusion fine-tune achieves the best average F1 among the merged generalist results, increasing mean F1 from 0.6150 to 0.6258. The existing positive generalist modifications are retained for comparison: the smooth denoised diffusion objective increases mean F1 to 0.6256, goal side-channel representation increases mean F1 to 0.6235, and reward-based rollout selection increases mean F1 to 0.6196.

Figure 2 visualizes the specialist F1 learning curve and the motion-weighted low-level diffusion sweep from the available evaluation CSVs. The generalist sweep is non-monotonic, with the best mean F1 at $\alpha = 3$, and the per-clip panel shows that the mean improvement still hides substantial clip-level variation.

4.5 COMPACT DIAGNOSTIC OUTCOMES

Negative and inconclusive diagnostics help interpret the positive results. Naive action smoothing and extra DDPM steps reduce F1 because contact timing is fragile. Per-hand residual gates are too coarse, song-level features do not add useful local information beyond future-note observations, and no-teacher RPIM recovery drifts away from the PianoMime interface. Exact values, CSV files, and artifact paths are preserved in the project assets and notes.

Table 4: Main generalist results on five selected unseen clips.

Method	Precision	Recall	F1	$\Delta F1$
Baseline	0.7000	0.6156	0.6150	—
Motion-weighted Low-level Diffusion Fine-tune	—	—	0.6258	+0.0108
Smooth Diffusion Objective	0.7010	0.6248	0.6256	+0.0105
Goal Representation	0.7064	0.6254	0.6235	+0.0085
Reward-based Rollout Selection	0.7117	0.6132	0.6196	+0.0046

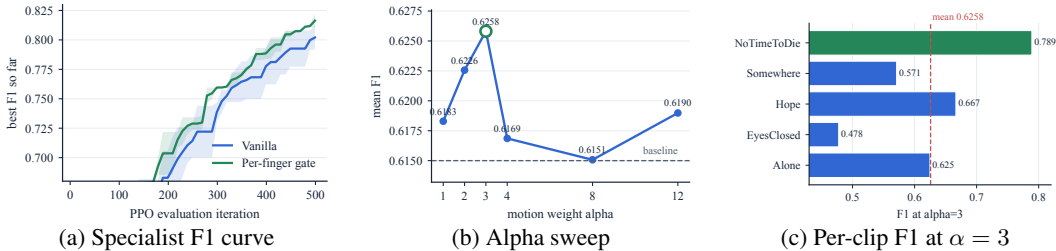


Figure 2: F1 curves and motion-weighted low-level diffusion fine-tuning. The specialist panel reports the F1 learning trend; the generalist sweep finds the best point at $\alpha = 3$, with mean F1 0.6258 against the merged-paper baseline 0.6150.

4.6 ADDITIONAL POSITIVE EVIDENCE

In the broader RPIM evaluation, aligned low-level replacement improves `Forester_1` from 0.6731 to 0.6986. Teacher-regularized recovery reaches about 0.6897 on `Forester_1`. These positive single-clip diagnostics indicate that external robot piano data can help after interface alignment. At the same time, the current five-clip RPIM averages remain below the official baseline, so we treat them as diagnostics rather than main positive results and keep the detailed numbers in the project assets and notes.

5 ARM-MOUNTED SHADOW HANDS

The standard RoboPianist and PianoMime setups use floating Shadow Hands whose wrist poses are not generated by physical robot arms. As a course bonus exploration, we built a MuJoCo / dm-control variant where two 6-DoF UR10-style arms carry the original RoboPianist Shadow Hands. The hand forearm translation degrees of freedom are removed, so the arms determine the wrist poses. We added stronger damping, gravity compensation, bounded actuator ranges, and implicit integration to keep the arm-hand system stable under contact.

Directly training the full arm-hand system from scratch was too difficult: the policy often failed to separate arm positioning from finger pressing, and arm motion created many wrong-key contacts. The effective route was reference based. We copied a floating-hand rollout, solved continuous UR10 IK frame by frame so the mounted wrists followed the original hands, optimized residual action sequences around this reference, and then distilled the optimized sequence with BC/Dagger. For `Happy_1`, the retargeting diagnostic reached 2.68 mm median wrist-position error and 6.69 mm 90th-percentile error, with 0.0392 rad 90th-percentile arm-joint change between adjacent frames.

Under the no-clamp metric, the mounted action sequence reaches F1 0.8073 on `Happy_8` with precision 0.7309 and recall 0.9016, while the floating-hand specialist reaches F1 0.9192. On `Happy_1`, the same mounted pipeline reaches F1 0.7451. Figure 3 illustrates the main failure mode: the arm can place the hand near the correct region, but wrist and palm inertia make lift-off slower, so neighboring fingers can keep contacting keys after the target transition. The arm is therefore not just a visual attachment; it changes exploration, stability, and contact precision.

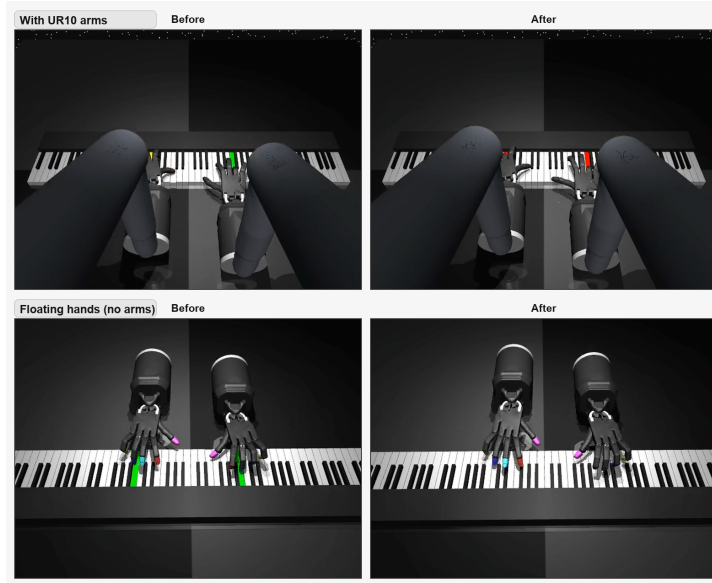


Figure 3: Qualitative effect of adding UR10 arms: delayed lift-off causes false-positive key contact compared with the floating-hand reference.

6 DISCUSSION AND CONCLUSION

The specialist experiments show that note onset frames deserve explicit attention. The combined method improves `Happy_8` from 0.8725 to 0.9192 F1, supporting onset-aware reward shaping and residual calibration. The per-finger gate diagnostic further suggests that useful residual control should be local to the finger and onset rather than coarse at the hand level.

The generalist experiments show that motion stability helps only when contact timing is preserved. Motion-weighted low-level diffusion fine-tuning gives the strongest merged result, increasing the five-clip average from 0.6150 to 0.6258; smooth denoised diffusion training remains a close positive comparison at 0.6256. In contrast, post-hoc smoothing and extra DDPM steps degrade timing. Goal side-channel conditioning and reward-based rollout selection also improve the five-clip average, respectively changing the training-time conditioning signal and inference-time selection rule.

The RP1M and UR10 experiments expose two interface problems. RP1M provides broader robot demonstrations, but direct replacement and no-teacher recovery can drift away from the PianoMime low-level action/state interface; teacher regularization and learned representation mapping are therefore necessary for stable transfer. Similarly, floating Shadow Hands hide the difficulty of placing the wrist with a physical arm. Reference-based retargeting, global F1 action-sequence optimization, and DAgger distillation gave a workable no-clamp arm-mounted result, but the remaining false positives show that realistic embodiment mainly hurts precision.

Overall, the experiments show that note onsets, contact-preserving motion objectives, local conditioning, representation alignment, and physical embodiment constraints are the key bottlenecks for robust dexterous piano playing.

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